
Dynamical System Discovery from Specifications and Partial Observations

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Abstract

Scientific model discovery is often framed as fitting ordinary differential equations to observed trajectories. In many applications, however, the desired model is defined not only by prediction error but also by a *task specification*: a natural-language description of what the model must support. In postprandial glucose modeling, for example, a meal-response simulator must represent how ingested carbohydrates enter the bloodstream over time; a post-meal decision-support model must distinguish meal absorption from delayed insulin action; and a physiology-oriented model must expose hepatic regulation. These task-critical mechanisms are often not measured, creating a mismatch between what the data observe and what the model must represent. We study this setting as *task-specified dynamical system discovery* under *specification-induced partial observability*.

We propose a framework that treats the state specification itself as an object of discovery. Given a task description and partially observed time series, our method jointly searches over interpretable state variables, analytic vector fields, and continuous parameters via maximum a posteriori (MAP) inference with data-fit and specification-grounding objectives. A proposer large language model (LLM) generates candidate state specifications and mechanism templates, a judge LLM scores whether the resulting model satisfies the task requirements, and a graph meta model converts proposed mechanisms into structured function libraries for efficient parameter fitting and pruning. Missing task-critical states are handled through observable-to-latent surrogate constructions inspired by delay embedding. On postprandial glucose–insulin benchmarks requiring progressively richer latent mechanisms, our method more reliably recovers task-critical structure than data-only and LLM-guided equation discovery baselines while maintaining competitive prediction accuracy.

1 Introduction

Scientific model discovery is often posed as equation discovery over a *given* state space: the learner is given trajectories of measured variables and searches for a vector field that predicts their evolution. This view underlies sparse regression [1], symbolic regression [2], neural ODEs [3], and recent LLM-guided equation discovery [4]. It is appropriate when the variables required by the target model are already measured, or can be constructed directly from measurements. In many scientific settings, however, **the target model is defined by the task it must support**. A postprandial glucose model for sensor testing must represent meal appearance; a model for debugging insulin-therapy response must distinguish meal-driven glucose input from delayed insulin action; and a physiological simulator may need to expose hepatic regulation or insulin secretion. These task-critical mechanisms are often absent from the measurements, so accurate prediction alone does not guarantee that the discovered equations are usable for the intended task.

Dynamical system discovery under specification-measurement mismatch

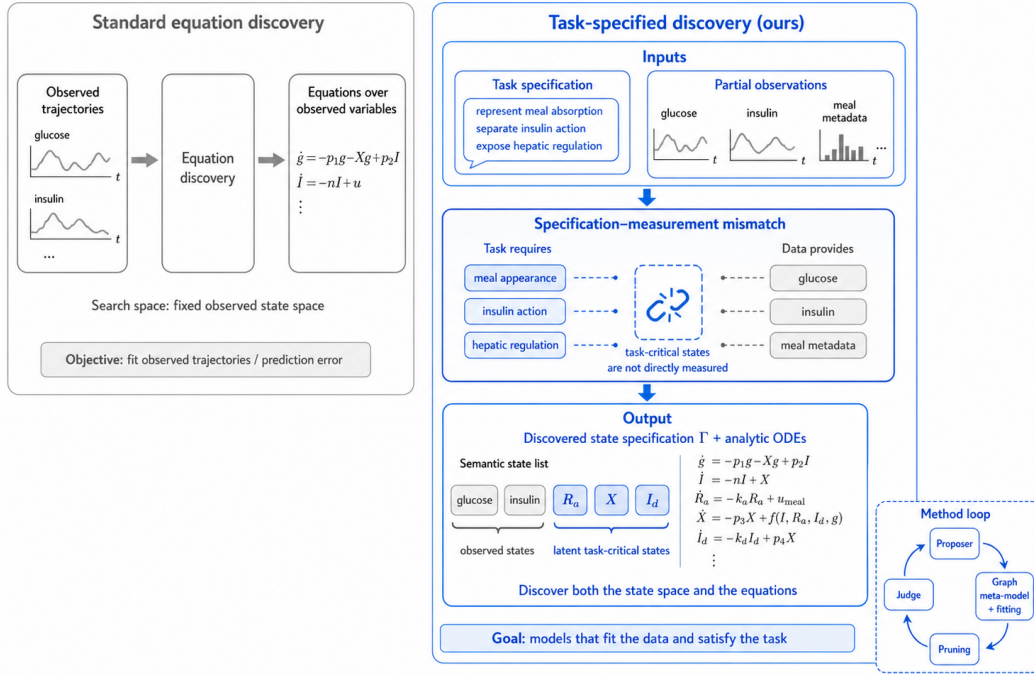


Figure 1: Comparison between standard equation discovery and task-specified discovery (**ours**), with components in existing LLM-guided equation discovery highlighted in gray and new components in our work in blue. We study a novel discovery regime, where the model must satisfy a *natural-language task specification*, and task-critical states may be absent from the *partially observed* data.

We therefore consider a *different* problem: discovering a dynamical model whose state space is not given in advance, but must be inferred from both a task specification and data. The input is a natural-language specification and partially observed time series; the output is an analytic dynamical system with a state representation sufficient for the specified task. This requires deciding which state variables should be present, which are observed or latent, and how latent task-critical mechanisms can be inferred from the available measurements. We call this setting **task-specified dynamical system discovery under specification-induced partial observability**. Unlike standard equation discovery, which searches for equations over a fixed observed state space, it requires joint discovery of state variables, latent state reconstruction, and governing equations.

This problem raises two difficulties that are largely absent from standard equation discovery. First, **state-specification ambiguity**: a task specification may indicate *which* mechanism to expose, but not *the state representation in which it should be encoded*. For example, a delayed regulatory effect could be represented by a single insulin-action state, a short chain of compartments, or another low-dimensional surrogate. Thus the state space is not a fixed input to discovery; it is part of what must be inferred. Second, **specification-induced partial observability**: the variables missing from the data are often exactly those required by the task. In postprandial glucose modeling, the task may require separating meal appearance, insulin-driven disposal, and delayed hepatic regulation, while the data contain only glucose, insulin, and meal metadata. A successful model must therefore construct not only task-critical latent states themselves but also the *reconstruction map from observed trajectories to latent states*. This makes discovery a joint problem of state space construction, latent mechanism recovery, and equation fitting.

Our key idea is to make the state space itself part of model discovery. To address *state-specification ambiguity*, we search over semantic state specifications, including both observed and latent variables with interpretable roles induced by the task specification. To address *specification-induced partial observability*, we require latent task-critical states to be connected to measured trajectories through explicit observation-to-latent reconstruction mechanisms, rather than introduced as unconstrained hidden dimensions. We formulate this as a maximum a posteriori (MAP) problem whose posterior

combines data fit with a specification-grounding prior. Operationally, a proposer large language model (LLM) generates candidate state specifications and analytic mechanism templates; continuous parameters are fit and pruned; and a judge LLM evaluates whether the resulting model satisfies a rubric induced by the task specification. Graph meta models provide the structured representation that makes this mixed discrete–continuous search tractable.

Contributions. ① We formalize *task-specified dynamical system discovery under specification-induced partial observability*, where the target ODE model is constrained jointly by observed trajectories and by a natural-language task specification (Sec. 3). ② We derive a MAP objective with an energy-based specification prior, enabling state specifications and vector fields to be scored by both trajectory fit and task compliance. ③ We introduce a graph meta model, together with Takens-inspired observation-to-latent constructions, that supports interpretable latent mechanisms and closed-form least-squares parameter fitting with pruning (Sec. 4). ④ We construct a glucose–insulin benchmark suite that varies task-required latent mechanisms and observability levels, and show that our method consistently recovers latent mechanisms, while existing data-only and LLM-guided baselines fail to do so (Sec. 5).

2 Related Works

Equation discovery over fixed state spaces. Sparse and symbolic equation discovery methods recover analytic dynamics by searching over candidate functions of *specified states*. SINDy identifies sparse governing equations from measurement data [1], while symbolic regression such as PySR broadens the functional search space through genetic or evolutionary search [2]. These methods are effective when the variables needed for the target model are measured or manually engineered. Their search space, however, is typically defined over the observed state and a candidate function library. They therefore do not address cases where the task specification requires additional interpretable states that are absent from the data.

Latent dynamical models. Neural state-space models (neural ODEs [3]) and latent variants of SINDy [5] address partial observability by learning hidden representations that improve prediction. This is complementary but different from our objective. The learned latent dimensions are typically not interpretable, are not required to correspond to task-specified mechanisms, or are not evaluated by whether they expose variables needed for downstream use. Our goal is not only to introduce hidden dimensions, but to discover interpretable latent states whose semantics are induced by the task.

Observability and delay embeddings. Classical observability theory and delay-coordinate embeddings study when hidden system states can be reconstructed from measurements [6]. These results motivate using observable histories or derivatives as surrogates for unmeasured states. They do not, however, determine which reconstructed coordinates should appear in a scientific model, nor whether those coordinates correspond to mechanisms required by a task. We use embedding ideas as constraints on identifiable latent-state proposals, while relying on the task specification to select semantically meaningful mechanisms.

Scientific priors and formal constraints. Other approaches incorporate scientific knowledge through formal constraints, axioms, invariances, units, or grammars [7, 8]. These methods move beyond unconstrained data fitting, but they usually assume that the relevant variables and constraints have already been formalized. We address an earlier formulation problem: the task is given in natural language, the required mechanisms may be implicit, and the corresponding state variables may be missing from the measurements.

LLM-guided equation discovery. Recent methods use LLMs to improve proposal generation and search in symbolic regression [4, 9] and dynamical system discovery [10, 11]. These works show that language models can improve equation search. Our setting differs in the role assigned to language: the natural-language specification is not merely background knowledge for proposing better equations, but a source of requirements defining which mechanisms the model must contain. Accordingly, our objective includes explicit specification grounding, and our search space includes semantic latent states beyond the observed variables. A detailed comparison with LLM-based methods is provided in Table 1 after we describe the problem formulation.

3 Problem Formulation

We consider the problem of formulating analytical dynamical systems from a natural-language task specification and observational time series. The input consists of (i) a *context* $\mathcal{C}$ describing the modeling task (e.g., “design an insulin replacement therapy”), and (ii) a *dataset* $\mathcal{D} = \{\mathbf{Y}_i\}_{i=1}^N$ of N trajectories, where each $\mathbf{Y}_i = [\mathbf{y}_i(t_{i,k})]_{k=1}^{T_i}$ are noisy measurements of the *observable states* $\mathbf{x}_{\text{obs}}(t)$ of the system taken at time stamps $\{t_{i,k}\}_{k=1}^{T_i}$:

$$\mathbf{y}_i(t_{i,k}) = \mathbf{x}_{\text{obs}}(t_{i,k}) + \boldsymbol{\varepsilon}(t_{i,k}) \in \mathbb{R}^{d_y}, \quad (1)$$

with $\boldsymbol{\varepsilon}(t_{i,k}) \in \mathbb{R}^{d_y}$ being the measurement noise.

Our goal is to discover an *analytical dynamical model*

$$\mathcal{M} = (\Gamma, f, \psi), \quad (2)$$

that “*matches*” both *context* and *data*. Here, Γ is *state specification*, defined as an ordered list $\{v_1, \dots, v_{d_x}\}$ of *state variable definitions*, where each v_i includes the semantic name (e.g., ‘insulin’, ‘glucose’) and its attributes (e.g., units). The vector field $f : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_x}$, together with its continuous parameters ψ , defines the governing dynamics

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{x}_{\text{obs}}(t) \\ \mathbf{x}_{\text{lat}}(t) \end{bmatrix} \in \mathbb{R}^{d_x}, \quad \dot{\mathbf{x}}(t) = f(\mathbf{x}(t); \psi), \quad (3)$$

where $\mathbf{x}(t) \in \mathbb{R}^{d_x}$ is the full state vector and $\mathbf{x}_{\text{lat}}(t)$ is the latent components.

Remark. The state specification Γ is *different from the state value* $\mathbf{x}$; it is a key component of the model that distinguishes our formulation from existing works: we explicitly include interpretable state representation Γ as an optimization variable. We write $\Gamma_{\text{obs}}(\mathcal{D})$ as the *observable state specification from the data*.

We formalize “matching context and data” as maximum a posteriori (MAP) inference:

$$\mathcal{M}^* = \arg \max_{\mathcal{M}} p(\mathcal{M} \mid \mathcal{C}, \mathcal{D}). \quad (4)$$

Assuming data $\mathcal{D}$ is generated by model $\mathcal{M}$, namely $\mathcal{D} \perp \mathcal{C} \mid \mathcal{M}$, the MAP inference becomes

$$\mathcal{M}^* = \arg \max_{\mathcal{M}} p(\mathcal{D} \mid \mathcal{M}) \cdot p(\mathcal{M} \mid \mathcal{C}). \quad (5)$$

Challenges Formulated and Example.

Consider the example of post-meal glucose–insulin modeling for insulin therapy design. **Prompt:** “Build an interpretable dynamical model suitable for designing an insulin replacement therapy; the model should represent insulin action and meal-induced glucose dynamics.” **Dataset:** plasma glucose time series from meal tests.

(C1) Context grounding $p(\mathcal{M} \mid \mathcal{C})$: the prompt implies semantic requirements (e.g., insulin-related state variables and mechanisms for insulin action) that must be enforced through the context-conditioned term $p(\mathcal{M} \mid \mathcal{C})$.

(C2) Incomplete state specification $|\text{support}[p(\Gamma \mid \mathcal{C})]| > 1$: the prompt does not uniquely determine the state representation, so the best representation Γ needs to be inferred (e.g., one- or two-compartment model with possible delayed insulin-action states).

(C3) Misalignment of context and data ($p[\Gamma_{\text{obs}}(\mathcal{D}) \mid \mathcal{C}] = 0$): variables required by the task (e.g., insulin action) may be missing from the dataset, forcing Γ to contain interpretable latent states with unknown initial values.

(C4) Mixed discrete-continuous search space (Γ, f, ψ) : jointly searching over discrete state representation Γ and functional forms f and continuous parameters ψ is a mixed discrete–continuous optimization, motivating representations that enable efficient inner-loop parameter estimation while an LLM explores the discrete state specification space.

We compare with LLM-SR and D3 within the framework of our problem formulation in Table 1.

Table 1: Detailed comparison with LLM-SR [4] and D3 [11] in terms of (1) whether the objective includes an explicit context-grounding term (2) whether the search space of state specification goes beyond the observed data, and (3) whether the algorithm efficiently handles the complex search space with an LLM in the loop.

	(1)	(2)	(3)	Objective	State Specification	Algorithm
LLM-SR	×	×	✓	$p(\mathcal{D} \mathcal{M})$ only	$\Gamma = \Gamma_{\text{obs}}(\mathcal{D})$	f : LLM, ψ : nonconvex
D3	×	×	×	$p(\mathcal{D} \mathcal{M})$ only	$\Gamma \subseteq \Gamma_{\text{obs}}(\mathcal{D})$	(Γ, f) : LLM, ψ : nonconvex
Ours	✓	✓	✓	$\underbrace{p(\mathcal{D} \mathcal{M})}_{\text{data-fitting}} \times \underbrace{p(\mathcal{M} \mathcal{C})}_{\text{spec-grounding}}$	$\Gamma \subseteq \Gamma_{\text{obs}}(\mathcal{D}) \cup \Gamma_{\text{lat}}$	(Γ, f) : LLM, ψ : closed-form

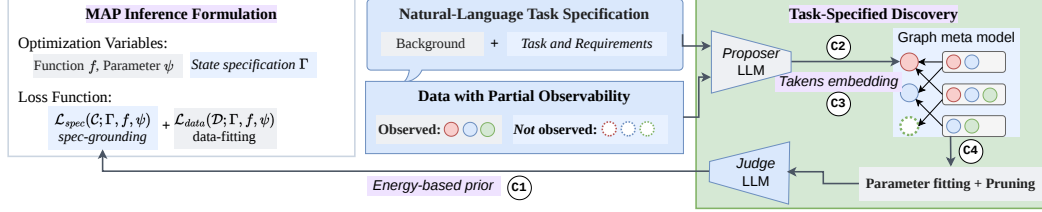


Figure 2: How the proposed method addresses the challenges in Section 3? To solve our MAP formulation with data-fitting and specification-grounding losses, a proposer LLM generates candidate state specifications (C2) and analytic mechanism templates inspired by Takers embedding (C3); a graph meta model provides the structured representation that results in closed-form solution to continuous parameters (C4); and a judge LLM gives spec-grounding loss as energy-based prior (C1).

4 Method

Our framework (Fig. 2) efficiently solves the MAP inference problem (5). First, the MAP problem is equivalent to minimizing the negative log-posterior

$$\min_{\Gamma, f, \psi} \underbrace{-\log p(\mathcal{D}|\Gamma, f, \psi)}_{\mathcal{L}_{\text{data}}, \text{MSE}} \underbrace{-\log p(\Gamma, f|\mathcal{C})}_{\lambda_{\text{spec}} \mathcal{L}_{\text{spec}}, \text{LLM score}} \underbrace{-\log p(\psi|\Gamma, f, \mathcal{C})}_{\lambda_{\text{reg}} \mathcal{L}_{\text{reg}}, \text{pruning}}. \quad (6)$$

The first term $\mathcal{L}_{\text{data}}(\Gamma, f, \psi; \mathcal{D})$ measures the data fit. The second term $\lambda_{\text{spec}} \mathcal{L}_{\text{spec}}$ is the *specification-grounding prior* scored by an LLM evaluator and serves as an energy-based prior $p(\Gamma, f|\mathcal{C}) \propto \exp[-\lambda_{\text{spec}} \mathcal{L}_{\text{spec}}(\Gamma, f; \mathcal{C})]$ [12]. The last term $\lambda_{\text{reg}} \mathcal{L}_{\text{reg}}$ is a sparsity-inducing regularizer to promote parsimonious analytical dynamics. Next, we discuss how to minimize these loss terms.

4.1 Data-Fitting Loss $\mathcal{L}_{\text{data}}(\Gamma, f, \psi; \mathcal{D})$ — Addressing (C3) (C4)

Under the widely adopted assumption of Gaussian noises, the data fit loss $\mathcal{L}_{\text{data}}(\Gamma, f, \psi; \mathcal{D})$ reduces to the MSE [13]

$$\frac{1}{\sum T_i} \sum_{i=1}^N \sum_{k=1}^{T_i} \|\mathbf{y}_i(t_{i,k}) - \hat{\mathbf{x}}_{i,\text{obs}}(t_{i,k}; \Gamma, f, \psi)\|_2^2, \quad (7)$$

where $\hat{\mathbf{x}}_{i,\text{obs}}(t_{i,k}; \Gamma, f, \psi)$ is the predicted observable states from the model (Γ, f, ψ) . We need to address two challenges: (1) the observable states $\hat{\mathbf{x}}_{i,\text{obs}}(t_{i,k}; \Gamma, f, \psi)$ depends on unknown latent variables $\mathbf{x}_{\text{lat}}$ ((C3)), and (2) the MSE is *non-convex* in continuous parameters ψ ((C4)).

Recovering Interpretable Latent Variables. Our approach to recover latent variables is theoretically grounded in the Takers embedding theorem [6]. The theorem establishes that under certain conditions, the dynamics can be fully reconstructed by constraining the latent states to be a deterministic embedding of the observable states

$$\mathbf{x}_{\text{lat}}(t) = h(\mathbf{x}_{\text{obs}}(t); \zeta). \quad (8)$$

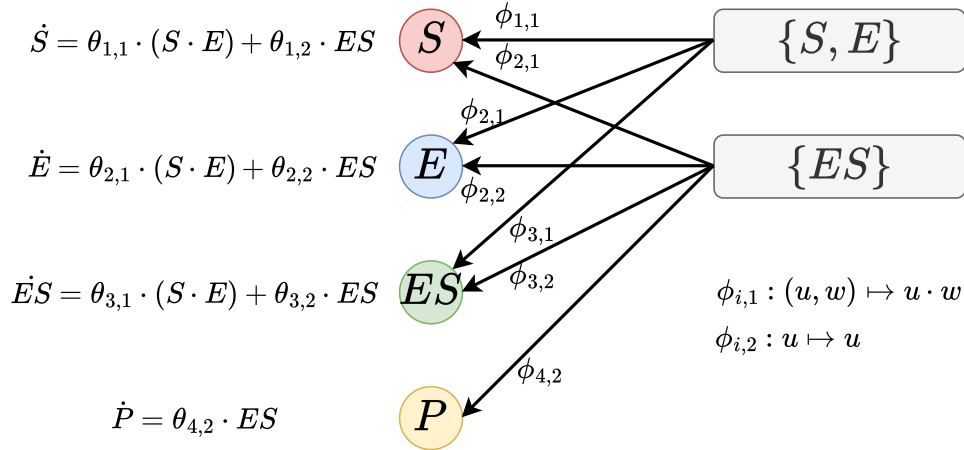


Figure 3: The graph meta model represents the nonlinear Michaelis-Menten kinetics [15, 16] as a linear combination of candidate functions.

142 The operational challenge in applying the Takens theorem is that there are infinite number of valid
 143 embeddings $h(\cdot; \zeta)$, and that the commonly used ones, namely derivatives of $\mathbf{x}_{\text{obs}}(t)$ up to certain
 144 order, may result in latent variables that are not interpretable. We leverage the LLM as a physics-
 145 informed generator that proposes functional forms $h(\cdot; \zeta)$ that result in physically meaningful $\mathbf{x}_{\text{lat}}$
 146 (e.g., proposing $v = \dot{x}$ or $E \propto \dot{x}^2$). Then we can calculate the observable states by

$$\mathbf{x}_{\text{obs}} = \hat{f}([\mathbf{x}_{\text{obs}}; h(\mathbf{x}_{\text{lat}}; \zeta)]; \psi). \quad (9)$$

147 **Dealing with Non-Convexity via Graph Meta Models.** Given the state specification Γ and vector
 148 field f , optimizing the continuous parameters ψ is challenging for nonlinear vector fields that result
 149 in non-convexity of the MSE in ψ . Therefore, we propose a novel *graph meta model* to represent a
 150 more general function library, so that the vector field can be expressed as a *linear combination* of
 151 candidate nonlinear functions.

152 **Definition 1** (Graph Meta Model). A graph meta model $\mathcal{G}$ is a bipartite *hypergraph* [14], with one
 153 partition of nodes being the state specification $\Gamma = \{v_1, \dots, v_{d_x}\}$ and the other partition include an
 154 arbitrary subset of Γ . An edge is a triple

$$(v \in \Gamma, S \subseteq \{1, \dots, d_x\}, \phi), \quad (10)$$

155 where S is an index set indicating the dependent state variables and $\phi(\mathbf{x}_S)$ is a function capturing the
 156 dependency.

157 For state x_i , the dynamics can be represented by

$$\dot{x}_i = \sum_{j \in \mathcal{E}_i} \theta_{i,j} \cdot \phi_{i,j}(\mathbf{x}_{S_{i,j}}), \quad (11)$$

158 where $\mathcal{E}_i$ is the set of edges connecting to v_i with the j th edge $(v_i, S_{i,j}, \phi_{i,j})$ weighted by $\theta_{i,j}$.

Remark. In a graph meta model, an edge can encode the dependency of one state on *multiple*
states simultaneously, crucial for modeling nonlinear dynamics as linear combination of candidate
 159 functions. Fig. 3 illustrates how the graph meta model represents nonlinear dynamics.

160 Given a graph meta model $\mathcal{G}$, the learnable parameters are the weights $\{\theta_{i,j}\}$, and the MSE is a
 161 quadratic function of the parameters. Hence, the optimal parameters have a closed-form solution. Let
 162 $\Phi_i \in \mathbb{R}^{T \times m_i}$ be the design matrix of candidate mechanisms for state x_i . Then we have

$$\theta_i^* = (\Phi_i^\top \Phi_i + \lambda I)^{-1} \Phi_i^\top \mathbf{y}_i, \quad (12)$$

163 where $\lambda > 0$ is a small number introduced for numerical stability.

164 4.2 Specification-Grounding Loss $\mathcal{L}_{\text{spec}}(\Gamma, f; \mathcal{C})$ — Addressing C1 C2 C3

We use a separate LLM as a *judge* that evaluates how the proposed state specification Γ and vector field f matches the context $\mathcal{C}$. The judge LLM returns a score $s_{\text{judge}}(\Gamma, f; \mathcal{C})$. Note that we do not interpret the judge LLM’s score as a calibrated/true probability $p(\Gamma, f | \mathcal{C})$. Instead, we treat it as a *surrogate compatibility score* between a candidate structure (Γ, f) and the context $\mathcal{C}$:

$$\text{Energy}_{\text{ctx}}(\Gamma, f; \mathcal{C}) \triangleq -\log s_{\text{judge}}(\Gamma, f; \mathcal{C}),$$

and convert it into an energy-based structural prior used in MAP inference:

$$p(\Gamma, f | \mathcal{C}) \propto \exp[-\lambda_{\text{ctx}} \cdot \text{Energy}_{\text{ctx}}(\Gamma, f; \mathcal{C})].$$

165 This is a standard energy-based modeling [12]: any positive scoring function can define an energy
166 and hence an unnormalized prior that biases the posterior toward structures satisfying constraints not
167 captured by data fit (e.g., “must include insulin” in an insulin-therapy task).

168 The original MAP inference (5) is equivalent to minimizing a weighted sum of the MSE and
169 the negative log of the LLM score. In practice, the exact weight cannot be determined. In our
170 implementation, we combine the MSE and the LLM score by their ratio $\frac{\mathcal{L}_{\text{data}}(\Gamma, f, \psi; \mathcal{D})}{s_{\text{judge}}(\Gamma, f; \mathcal{C})}$. The key
171 advantage is that it requires no hyperparameter (i.e., weight) and is invariant to the observation noise
172 power. Importantly, it captures the same trend as the original loss function: when the score s_{judge} is
173 close to zero, both $\text{MSE}/s_{\text{judge}}$ and $-\log p(\Gamma, f | \mathcal{C})$ result in extremely high loss. In other words,
174 the state specification Γ and vector field f that is not context grounded, measured by a near-zero
175 score s_{judge} , will have arbitrarily large loss.

176 4.3 Regularization Loss $\mathcal{L}_{\text{reg}}(\psi)$ — Promoting Sparsity by Pruning

177 The sparsity loss $\mathcal{L}_{\text{reg}}(\psi)$ requires calculating the prior $p(\psi | \Gamma, f, \mathcal{C})$, which may be infeasible. We
178 minimize this loss term by pruning, which is essentially estimating the prior through data via non-
179 parametric bootstrap [17]. Specifically, a collection of n bootstrap datasets $\mathcal{D}_1, \dots, \mathcal{D}_n$ is generated
180 by resampling the original dataset with replacement, and a set of estimated coefficients $\{\theta_{i,j}\}$ are
181 calculated by least-squares under each bootstrap sample.

182 Since graph meta model represents the model as linear combination of functions, we can derive an
183 analytical expression for the negative log-likelihood of the coefficient being zero [18]. We define
184 it as the importance score, and prune the terms whose coefficient has a significance score below a
185 threshold. The detailed derivation is in Appendix B.

186 5 Experiments

187 We design the experiments to answer two key questions:

188 **Q1. Mechanism recovery:** Does our method recover the task-critical latent mechanisms required by
189 the specification?

190 **Q2. Specification–measurement mismatch:** Does the advantage of specification-grounded discov-
191 ery increase as fewer task-relevant variables are observed?

192 In addition, we will evaluate the standard metric of prediction accuracy, measured by the MSE.

193 5.1 Benchmarks, Baselines, and Metrics

194 We build 12 glucose–insulin benchmark instances based on [19] (summarized in Table 2). The
195 four tasks require progressively richer latent mechanisms: meal appearance, insulin action, delayed
196 hepatic regulation, and full flux decomposition. Within each task, the easy/medium/hard tiers reduce
197 the number of observed variables revealed to the learner. Detailed description is in Appendix C.

198 **Baselines.** We have three baselines as representatives of different classes of methods.

- 199 • **SINDy** [1]: represent purely data-driven methods that fit the trajectories with a given
200 function library;

	Task 1			Task 2			Task 3			Task 4		
	E	M	H	E	M	H	E	M	H	E	M	H
Task Requirements	Meal appearance			Insulin action			Hepatic delay			Full flux portrait		
# Latent Mechanisms	1	1	1	2	2	2	3	3	3	4	4	4
# Observed Variables	5	2	1	4	3	2	4	3	2	6	4	2
# Task-Critical Variables	1	1	1	1	1	1	1	1	1	4	4	4

Table 2: Summary of the 12 benchmark instances consisting of four tasks, each of which has easy (E), medium (M), and hard (H) tiers. For each benchmark, we list the task requirements, the number of latent mechanisms required, the number of observed variables (i.e., time-series variables revealed to the learner), and the number of task-critical variables.

- **PySR** [2]: represents purely data-driven methods that can automatically find the function forms to best fit the data;
- **D3** [11]: represents LLM-guided methods that can take text as input, but do not consider task specification.

Metrics. We evaluate the methods on two key metrics. The first is **state-specification success**. A run is counted as a state-specification success if the discovered model contains an explicit interpretable state or process for *every* mechanism required by the task, and each mechanism is generated by the model equations. The second is **latent mechanism recovery**. When ground-truth hidden trajectories are available from the simulator, we evaluate recovery using the MSE of hidden trajectories. For baselines that do not output a required hidden mechanism, we mark the mechanism recovery as failed and do not assign a trajectory error for that mechanism.

In addition to the two key metrics, we also report commonly used metrics of *MSE of observed trajectories* and the *number of nonzero terms* in the discovered ODEs.

5.2 Main Results

We summarize results on all 12 benchmarks in Table 4 (Appendix D). Our method is the only one that consistently recovers the required latent mechanisms on all the benchmarks. It also discovers equations with fewer terms across all benchmarks. We can also see that prediction error alone does not characterize success in task-specified model discovery. Several baselines achieve reasonable or best observed trajectory MSEs on easier instances, but fail to introduce the required task-critical mechanisms. Our method maintains competitive observed trajectory errors on each tasks and improved observed trajectory errors on the hardest tasks.

5.3 Case Study: Prediction Is Not Mechanistic Recovery

We next inspect the hardest instance, *T4-H: Flux Portrait Recovery under Partial Observability*. In this setting, the learner observes only glucose, insulin, and meal metadata, while the task requires a mechanistic decomposition into meal appearance, endogenous production, utilization, and insulin secretion. This benchmark therefore directly tests specification-induced partial observability: the task-critical mechanisms are not measured, but they must be present in the model for it to satisfy the specification.

Table 4 shows why prediction error alone is insufficient. On this benchmark, PySR obtains the lowest observed trajectory error, because it is actually relatively easy to track two trajectories. However, these baselines do not introduce the required flux mechanisms; their specification success is therefore zero and hidden-mechanism recovery is not applicable. In contrast, our method explicitly proposes latent process variables corresponding to the required flux portrait, achieving hidden trajectory MSE of 0.464 ± 0.067 , while maintaining a competitive observed trajectory fit. Thus, the case illustrates the central distinction of our setting: a model can fit the observed glucose and insulin trajectories while still being unusable for the task because it explains the data in the wrong state space.

Details on the discovered models are in Appendix E.

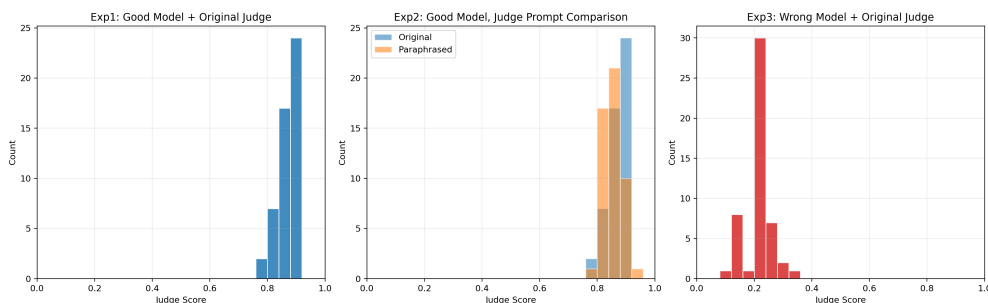


Figure 4: Consistency of the scores by the Judge LLM. **Left:** histogram of scores for a correct model; **Middle:** histogram of scores under paraphrased prompts; **Right:** histogram of scores for a wrong model.

5.4 Consistency of Scores by the Judge LLM

We assess the consistency of the judge score in the three aspects: **consistency for the same prompt**, where we evaluate the same model multiple times; **consistency of judges**, where we use different versions of judge prompts (same prompt paraphrased differently); **sanity checks**, where we compare the judge scores for the correct model (i.e., correct variables and function forms) and wrong models (e.g., models that miss a required variable and/or have incorrect function forms).

Fig. 4 shows that the LLM judge give consistently high scores to the correct model, distinctively low scores to wrong models, and consistent scores under different versions of prompts.

6 Conclusion and Discussion

We introduced *task-specified dynamical system discovery under specification-induced partial observability*, a setting in which the desired mechanistic model is defined not only by observed trajectories but also by a natural-language task specification. In this regime, accurate prediction is not sufficient: the model must be expressed in a state space that contains the mechanisms required by the task, even when those mechanisms are not directly measured. This creates two coupled challenges: state-specification ambiguity and specification-induced partial observability.

Our framework addresses these challenges by treating the state specification itself as an object of discovery. We formulate model discovery as a MAP inference problem that combines data fit with specification grounding. A proposer LLM searches over semantic state specifications and graph meta-models, continuous parameters are fit and pruned efficiently, and a judge LLM evaluates whether the resulting model satisfies task-induced requirements. To bridge from observations to unmeasured task-critical states, we use observation-to-latent surrogate constructions inspired by delay embedding, while preserving semantic interpretability.

Experiments on postprandial glucose–insulin benchmarks show that this distinction matters. Baselines can sometimes fit observed glucose or insulin trajectories, but often fail to recover the mechanisms required by the task, such as meal appearance, insulin action, delayed hepatic regulation, or the full flux portrait. Our method reliably recovers these task-critical mechanisms while maintaining competitive prediction accuracy.

Limitations. The current benchmarks focus on moderate-dimensional ODE systems, leaving scalability to larger biochemical or physiological networks as future work. Moreover, our treatment of specification grounding uses LLM-based scoring as a practical surrogate; further work may be needed to calibrate such scores and combine them with formal domain constraints when available. More broadly, we hope this work encourages a shift from trajectory-only equation discovery toward specification-grounded discovery of mechanistic models whose state spaces are aligned with the scientific tasks they are meant to support.

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321 A Extended Related Works

Table 3: Detailed comparison with existing works on symbolic regression and dynamical system discovery.

Method Family	Input	Task Specification	Partial Observability	Computation Challenge	Interpretability Challenge
Sparse Equation Discovery (SINDy [20])	data	—	×	N/A	N/A
PySR	data	—	×	N/A	N/A
AI-Hilbert	data	—	×	×	×
D3, LLM-ODE	text + data	×	×	×	×
Our Work	text + data	✓	✓	✓	✓

322 B Deriving Importance Score in Pruning

323 B.1 Details on Pruning

324 The pruning phase addresses the issue of over-parameterization in the initial Graph Meta Model
 325 (GMM) by enforcing physical parsimony through a rigorous stability analysis. The core objective
 326 is to filter out "spurious" terms — mathematical expressions that may fit the training noise but lack
 327 predictive validity — by analyzing the stability of their coefficients across perturbed datasets.

328 This is achieved via a non-parametric bootstrap approach [17], where a collection of n bootstrap
 329 datasets $\mathcal{D}_1, \dots, \mathcal{D}_n$ is generated by resampling the original snapshots with replacement. For each
 330 candidate term in the library, this process yields a set of estimated coefficients across bootstrap
 331 samples. We define the empirical probability density function for the coefficient c associated with
 332 the i -th term as $\rho_{c_i}(c) = \frac{1}{N} \sum_j \delta(c - c_i^{(\mathcal{G}_j)})$, where $c_i^{(\mathcal{G}_j)}$ is the coefficient value obtained in the j -th
 333 bootstrap realization. To robustly quantify the term's significance, the method employs a continuous
 334 approximation $\tilde{\rho}_{c_i}$ of this empirical density using a Gaussian distribution $\mathcal{N}(\hat{\mu}(c_i), \hat{\sigma}(c_i)^2)$, a model-
 335 ing choice justified by the asymptotic properties of the estimator. We motivate this approximation in
 336 Appendix B

337 Building on this probabilistic characterization, we formulate an *importance score* $s(c_i)$ to discriminate
 338 between consistent physical signals and noise. A library term is deemed spurious if its coefficient
 339 probability mass is significantly concentrated near zero. We quantify this via the negative log-
 340 likelihood of the coefficient being zero, $s(c_i) = -\log \tilde{\rho}_{c_i}(0)$, which serves as a proxy for the
 341 probability $\mathbb{P}[c_i \in (-\epsilon, \epsilon)]$ ¹. The final sparse model structure is determined through an iterative
 342 thresholding procedure where a set of candidate thresholds $\mathbf{t}$ is derived from the sorted importance
 343 scores. For each threshold $\tau \in \mathbf{t}$, the model is restricted to the active index set $\mathcal{I}_\tau = \{i \mid s(c_i) \geq \tau\}$
 344 and the parameters are refitted to maximize the likelihood on the reduced support. The optimal
 345 dynamical law is selected by minimizing the generalization error on a validation set across these
 346 candidate structures, and when multiple global optima exist the selection.

347 Building on this Gaussian model, the algorithm defines an *importance score* $s(c_i)$ to discern signal
 348 from noise. We define a "spurious" term as one whose coefficient has a high probability of being
 349 effectively zero (contained within a narrow interval $(-\epsilon, \epsilon)$). The probability mass in this interval is
 350 approximated by $2\epsilon \tilde{\rho}_{c_i}(0)$. To prioritize terms that are consistently non-zero, the importance score is
 351 defined as the negative log-likelihood [18] of the coefficient being zero:

$$s(c_i) = -\log \tilde{\rho}_{c_i}(0) = \frac{1}{2} \log(2\pi \hat{\sigma}(c_i)^2 + \eta) + \frac{\hat{\mu}(c_i)^2}{2\hat{\sigma}(c_i)^2 + \eta} \quad (13)$$

352 where η is a small constant introduced for numerical stability (e.g. $\eta = 10^{-8}$). The final model
 353 selection is performed via an iterative thresholding algorithm. The method sorts the computed scores
 354 s_i to generate a set of candidate thresholds $\mathbf{t}$. For each threshold $\tau \in \mathbf{t}$, a candidate model is
 355 constructed by restricting the GMM to the active set $\mathcal{I}_\tau = \{i \mid s_i \geq \tau\}$ and refitting the parameters
 356 Θ to maximize the likelihood on this subset. The optimal structure is then selected by minimizing the
 357 validation error over these candidate models, ensuring the final dynamical law balances sparsity with
 358 predictive accuracy.

¹For small enough ϵ the probability can be approximated as proportional to its density

B.2 Gaussian Approximation of Bootstrap Coefficients

The Gaussian approximation of bootstrap realizations of coefficients relies on the functional analytical view of the Ordinary Least Squares (OLS) estimator.

In particular, consider a bootstrap dataset $\mathcal{D}_{bootstrap}$. Such dataset is completely characterized by two objects:

1. The original dataset $\mathcal{D}$, where samples are drawn
2. The number of times w_k the sample k -th is extracted from the original dataset and appears in the bootstrap realization

In this framework, the trivial case $\mathbf{w} = \mathbf{1}$ generalizes classical coefficient reconstruction on the observed dataset. Therefore, we define the vector of coefficients as a function of the bootstrap dataset

$$\mathbf{c} := \mathbf{c}_{\mathcal{D}}(\mathbf{w}) \quad (14)$$

and, for the least square fitting, we note that such function is continuous and differentiable almost everywhere with respect to $\mathbf{w} \in \mathbb{R}^N$. This allows us to study a first order expansion of each coefficient function as, as

$$c_i(\mathbf{w} + \Delta \mathbf{w}) \approx c_i(\mathbf{w}) + \Delta \mathbf{w}^\top \nabla c_i(\mathbf{w}) \quad (15)$$

enabling us to express the effect of perturbations in each sample count on the final coefficients.

We also highlight that in the construction of bootstrap datasets the vector $\mathbf{w}$ is a random vector. In fact, it is straightforward to observe that the bootstrapping procedure yields

$$\mathbf{w} = \sum_i \mathbf{u}_i, \quad (16)$$

where $\mathbf{u}_1, \dots, \mathbf{u}_N$ is a collection of identically distributed random vectors, uniform over the canonical basis $\{\mathbf{e}_i\}_i$. Hence, the k -th weight can be studied as

$$\begin{aligned} w_k &= \mathbf{e}_k^\top \mathbf{w} \\ &= \mathbf{e}_k^\top \sum_i \mathbf{u}_i \\ &= \sum_i \mathbf{e}_k^\top \mathbf{u}_i \\ &= \sum_i b_i^{(k)} \end{aligned} \quad (17)$$

where $\{b_i^{(k)}\}_i$ is a collection of Bernoulli random variables of parameter N^{-1} , hence the marginal distribution of each weight is a binomial distribution. As an approximated proxy of independence we can also study their correlation as

$$\begin{aligned} \mathbb{E}[w_k w_j] - \mathbb{E}[w_k] \mathbb{E}[w_j] &= \mathbf{e}_k^\top (\mathbb{E}[\mathbf{w} \mathbf{w}^\top] - \mathbb{E}[\mathbf{w}] \mathbb{E}[\mathbf{w}^\top]) \mathbf{e}_j \\ &= \sum_{i\ell} \mathbf{e}_k^\top (\mathbb{E}[\mathbf{u}_i \mathbf{u}_\ell^\top] - \mathbb{E}[\mathbf{u}_i] \mathbb{E}[\mathbf{u}_\ell^\top]) \mathbf{e}_j \\ &= \sum_i \mathbf{e}_k^\top \left(\frac{1}{N} \mathbf{I} - \frac{1}{N^2} \mathbf{1} \mathbf{1}^\top \right) \mathbf{e}_j \\ &= \mathbf{e}_k^\top \left(\mathbf{I} - \frac{1}{N} \mathbf{1} \mathbf{1}^\top \right) \mathbf{e}_j \end{aligned} \quad (18)$$

discovering that weights have negative correlation, that vanishes as $\mathcal{O}(1/N)$ as the dataset size grows. Hence, we can study the coefficient function as the linear combination of weakly correlated random variables

$$\begin{aligned} c_i(\mathbf{w}) &= c_i(\mathbf{1} + (\mathbf{w} - \mathbf{1})) \\ &\approx c_i(\mathbf{1}) + \sum_k (w_k - 1) \partial_{w_k} c_i(\mathbf{1}) \end{aligned} \quad (19)$$

which justifies the Gaussian approximation for the distribution of coefficients, by means of CLT.

C Detailed Description of Benchmarks and Prompts

We evaluate our framework on a benchmark suite that is carefully designed to evaluate the capability of formulating partially observed dynamical systems under misaligned text and data.

C.1 Glucose-Insulin Benchmarks

We use the glucose-insulin model in [19], which characterizes the glucose-insulin regulation after a meal. The entire system is divided into the glucose subsystem and the insulin subsystem, each of which is described through ODEs and derived algebraic equations.

The full dynamics can be completely determined by a system of first-order ODEs with the following 12 states:

- G_p : Glucose mass in plasma and rapidly equilibrating tissues (mg/dL);
- G_t : Glucose mass in slowly equilibrating tissues (mg/dL);
- Q_{sto1} : Glucose in the stomach solid phase (mg);
- Q_{sto2} : Glucose in the stomach liquid phase (mg);
- Q_{gut} : Glucose in the intestine (mg);
- I_p : Insulin mass in plasma (pmol/L);
- I_l : Insulin mass in liver (pmol/L);
- I_d, I_1 : Delayed insulin signals realized with a chain of two compartments (pmol/L);
- I_{po} : Amount of insulin in the portal vein (pmol/kg);
- X : Insulin in the interstitial fluid (pmol/L);
- Y : Auxiliary insulin secretion state in β -cells (pmol/kg/min).

The ODEs governing the dynamics are as follows.

Glucose Subsystem:

$$\dot{G}_p(t) = EGP(t) + Ra(t) - U_{ii}(t) - E(t) - k_1 \cdot G_p(t) + k_2 \cdot G_t(t), \quad (20)$$

$$\dot{G}_t(t) = -U_{id}(t) + k_1 \cdot G_p(t) - k_2 \cdot G_t(t), \quad (21)$$

$$\dot{Q}_{sto1}(t) = -k_{gri} \cdot Q_{sto1}(t) + D \cdot \delta(t), \quad (22)$$

$$\dot{Q}_{sto2}(t) = -k_{empt} (Q_{sto1}(t) + Q_{sto2}(t)) \cdot Q_{sto2}(t) + k_{gri} \cdot Q_{sto1}(t), \quad (23)$$

$$\dot{Q}_{gut}(t) = -k_{abs} \cdot Q_{gut}(t) + k_{empt} (Q_{sto1}(t) + Q_{sto2}(t)) \cdot Q_{sto2}(t). \quad (24)$$

Insulin Subsystem:

$$\dot{I}_l(t) = -(m_1 + m_3(t)) \cdot I_l(t) + m_2 \cdot I_p(t) + S(t), \quad (25)$$

$$\dot{I}_p(t) = -(m_2 + m_4) \cdot I_p(t) + m_1 \cdot I_l(t), \quad (26)$$

$$\dot{I}_{po}(t) = -\gamma \cdot I_{po}(t) + S_{po}(t), \quad (27)$$

$$\dot{Y}(t) = \begin{cases} -\alpha \cdot [Y(t) - \beta \cdot (G(t) - h)] & \text{if } \beta \cdot (G(t) - h) \geq -S_b \\ -\alpha \cdot Y(t) - \alpha \cdot S_b & \text{if } \beta \cdot (G(t) - h) < -S_b \end{cases}, \quad (28)$$

$$\dot{I}_1(t) = -k_i \cdot [I_1(t) - I(t)], \quad (29)$$

$$\dot{I}_d(t) = -k_i \cdot [I_d(t) - I_1(t)], \quad (30)$$

$$\dot{X}(t) = -p_{2U} \cdot X(t) + p_{2U} \cdot (I(t) - I_b). \quad (31)$$

Note that the right-hand sides contain variables that are not among the 12 states, but can be derived from the state variables. Specifically, $EGP(t)$ is the endogenous glucose production (mg/kg/min)

$$EGP(t) = k_{p1} - k_{p2} \cdot G_p(t) - k_{p3} \cdot I_d(t) - k_{p4} \cdot I_{po}(t), \quad (32)$$

$Ra(t)$ is the glucose rate of appearance

$$Ra(t) = \frac{f \cdot k_{abs} \cdot Q_{gut}(t)}{BW}, \quad (33)$$

where f is the fraction of intestinal absorption and BW is the body weight, $U_{ii}(t)$ is the insulin-independent glucose utilization (a constant in this model)

$$U_{ii}(t) = F_{cns}, \quad (34)$$

$U_{id}(t)$ is the insulin-dependent glucose utilization

$$U_{id}(t) = \frac{[V_{m0} + V_{mx} \cdot X(t)] \cdot G_t(t)}{K_m + G_t(t)}, \quad (35)$$

$E(t)$ is the glucose renal excretion (mg/kg/min)

$$E(t) = k_{e1} \cdot \max\{G_p(t) - k_{e2}, 0\} \quad (36)$$

$S(t)$ is the insulin secretion

$$S(t) = \beta \cdot I_{po}(t), \quad (37)$$

$S_{po}(t)$ is the secretion in the portal vein

$$S_{po}(t) = Y(t) + K \cdot \max\{\dot{G}(t), 0\} + S_b, \quad (38)$$

$G(t)$ is the plasma glucose concentration (mg/dl)

$$G(t) = \frac{G_p(t)}{V_G}, \quad (39)$$

and $I(t)$ is the plasma insulin concentration (pmol/l)

$$I(t) = \frac{I_p(t)}{V_I}. \quad (40)$$

C.2 Benchmarks

Based on [19], we build four benchmarks with increasing difficulty levels by requiring progressively more coupled and confounded latent structures. In the first benchmark, the task requires one meal absorption latent variable. We add a requirement insulin-effect latent variable in the second benchmark, then a distinct hepatic (delayed) regulation pathway in the third benchmark, and finally the full multi-process flux portrait of the glucose-insulin process.

Within each benchmark, we also have three tiers of difficulty levels by controlling how much of the latent structure is revealed:

- Hard: only routinely available signals (typically plasma glucose concentration $G(t)$ and sometimes plasma insulin concentration $I(t)$).
- Medium: additionally reveal some tracer-derived fluxes from the database (e.g., $Ra(t)$, $EGP(t)$, $U(t)$, $S(t)$).
- Easy: reveal almost all measured concentration and flux variables, but still require latent internal states that are never measured.

C.2.1 Benchmark 1: Meal Appearance Recovery

Applications and use cases. This benchmark focuses on the glucose-insulin module only. Example use cases include testing of glucose sensors and characterization of meal absorption.

Equations to recover.

$$\dot{G}_p(t) = EGP(t) + Ra(t) - U_{ii}(t) - E(t) - k_1 \cdot G_p(t) + k_2 \cdot G_t(t). \quad (41)$$

Required unobservable variables. In these use cases, we require the glucose rate of appearance $Ra(t)$, the primary driver of the post-meal glucose rise, to correctly attribute the glucose rise to the meal and the endogenous production changes. However, the glucose rate of appearance is often unobservable.

428 **Observables.** We observe the glucose concentration $G(t)$ and the meal metadata (timing and
429 amount). We may observe additional variables in the easy and medium levels of this benchmark.

- 430 • Easy: in addition to $G(t)$ and meal timing and amount, we observe all non-meal contributions
431 to $\dot{G}_p(t)$, namely $EGP(t), U_{ii}(t), E(t), G_t(t)$.
- 432 • Medium: in addition to $G(t)$ and meal timing and amount, we observe major non-GI fluxes
433 $EGP(t), U_{ii}(t)$.
- 434 • Hard: we observe no additional variables, because $G(t)$ and meal timing and amount are the
435 only ones routinely measured in clinical practices.

436 **Proposer Prompt.**

437 *A. Domain context.* You are building a mechanistic, analytic dynamical model for postprandial
438 glucose simulation to support glucose-sensor evaluation and decision-support testing. After meal
439 ingestion, plasma glucose rises and falls over time due to physiological processes that include meal
440 absorption and glucose regulation.

441 *B. Requirements.*

- 442 • Propose a parsimonious continuous-time ODE model that can reproduce the observed
443 postprandial glucose trajectory given meal information.
- 444 • Your model must be able to explain how the meal influences plasma glucose over time (i.e.,
445 not an instantaneous effect). The latent mechanism should be causal and driven by meal
446 timing and amount.
- 447 • The model must be interpretable: provide meaningful names for states and explain each
448 state’s role.
- 449 • The model must be analytic, namely closed-form expressions.

450 *C. Constraints and plausibility check.*

- 451 • Nonnegativity: physiological quantities and latent states should remain nonnegative.
- 452 • Meal causality: meal influence should start at/after meal time, not before.
- 453 • Transient response: meal effect should rise and then decay; no unrealistic oscillations.
- 454 • Basal plausibility: before meal, the system should be at/near steady state (or you must state
455 assumptions/initialization).

456 *D. Data columns available.*

- 457 • In the easy level, we have the glucose concentration $G(t)$, the meal metadata (timing and
458 amount), and $EGP(t), U_{ii}(t), E(t), G_t(t)$.
- 459 • In the medium level, we have the glucose concentration $G(t)$, the meal metadata (timing
460 and amount), and $EGP(t), U_{ii}(t)$.
- 461 • In the hard level, we have the glucose concentration $G(t)$ and the meal metadata (timing
462 and amount).

463 *E. Required output format.*

464 Return, in order:

- 465 • State list: name, unit (if known), interpretation.
- 466 • ODEs: one equation per state.
- 467 • Meal input encoding: how meal time/amount enters the model.
- 468 • Observation mapping: how predicted glucose is computed from states.
- 469 • Sanity-check procedure: how you verify nonnegativity, causality, and transient response.

C.2.2 Benchmark 2: Absorption-Action Disentanglement

Applications and use cases. This benchmark targets mechanism-aware postprandial interpretation and decision-support debugging: given glucose responses after a meal, one often needs to distinguish whether a glucose excursion is driven primarily by meal absorption or by insufficient insulin effect on glucose disposal. This benchmark therefore extends Benchmark 1 by requiring an additional insulin-effect latent mechanism.

Equations to recover.

$$\dot{G}_p(t) = EGP(t) + Ra(t) - U_{ii}(t) - E(t) - k_1 \cdot G_p(t) + k_2 \cdot G_t(t), \quad (42)$$

$$\dot{G}_t(t) = -U_{id}(t) + k_1 \cdot G_p(t) - k_2 \cdot G_t(t), \quad (43)$$

$$U_{id}(t) = \frac{[V_{m0} + V_{mx} \cdot X(t)] \cdot G_t(t)}{K_m + G_t(t)}, \quad (44)$$

$$U(t) = U_{ii}(t) + U_{id}(t), \quad (45)$$

$$\dot{X}(t) = -p_{2U} \cdot X(t) + p_{2U} \cdot (I(t) - I_b). \quad (46)$$

Required unobservable variables. In these use cases, we require:

- glucose rate of appearance $Ra(t)$ to correctly attribute the post-meal glucose rise to the meal absorption process, and
- an insulin-effect latent that mediates insulin-dependent disposal (in the ground truth model, this is the interstitial/remote insulin action state $X(t)$, which governs $U_{id}(t)$).

Observables. We observe the glucose concentration $G(t)$, the meal metadata (timing and amount), and additionally insulin concentration $I(t)$. We may observe additional variables in the easy and medium tiers.

- Easy: in addition to $G(t), I(t)$ and meal timing and amount, we observe all non-meal contributions to $\dot{G}_p(t)$ and the major utilization terms needed to isolate insulin action, namely $EGP(t), U_{ii}(t), E(t), G_t(t), U(t)$.
- Medium: in addition to $G(t), I(t)$ and meal timing and amount, we observe the total utilization $U(t)$ and the insulin-independent utilization $U_{ii}(t)$ (a constant).
- Hard: we observe no additional variables, because $G(t), I(t)$ and meal timing and amount are the only ones routinely measured in clinical practices.

Proposer Prompt.

A. Domain context. You are building a mechanistic, analytic dynamical model for postprandial glucose simulation to support decision-support testing. After meal ingestion, glucose rises due to meal absorption and later decreases due to insulin-mediated disposal. Insulin’s effect on disposal is delayed and persistent relative to plasma insulin.

B. Requirements.

- Propose a parsimonious continuous-time ODE model that can reproduce the observed postprandial glucose trajectory given meal information, and insulin trajectory.
- Your model must be able to provide physiological interpretation that mediates the effect of insulin on glucose disposal over time.
- The model must be interpretable: provide meaningful names for states and explain each state’s role.
- The model must be analytic, namely closed-form expressions.

C. Constraints and plausibility check.

- Nonnegativity: physiological quantities and latent states should remain nonnegative.
- Meal causality: meal influence should start at/after meal time, not before.

- Transient response: meal effect should rise and then decay; no unrealistic oscillations.
- Basal plausibility: before meal, the system should be at/near steady state (or you must state assumptions/initialization).

D. Data columns available.

- In the easy level, we have the glucose concentration $G(t)$, the insulin concentration $I(t)$, the meal metadata (timing and amount), and $EGP(t)$, $U_{ii}(t)$, $E(t)$, $G_t(t)$.
- In the medium level, we have the glucose concentration $G(t)$, the insulin concentration $I(t)$, the meal metadata (timing and amount), the total utilization $U(t)$ and the insulin-independent utilization $U_{ii}(t)$ (a constant).
- In the hard level, we have the glucose concentration $G(t)$, the insulin concentration $I(t)$, and the meal metadata (timing and amount).

E. Required output format.

Return, in order:

- State list: name, unit (if known), interpretation.
- ODEs: one equation per state.
- Meal input encoding: how meal time/amount enters the model.
- Observation mapping: how predicted glucose is computed from states.
- Sanity-check procedure: how you verify nonnegativity, causality, and transient response.

C.2.3 Benchmark 3: Delayed Hepatic Regulation Recovery

Applications and use cases. This benchmark targets mechanistic physiology modeling in which the model must capture not only meal absorption and peripheral disposal but also the suppression of endogenous glucose production (EGP) after a meal. The benchmark adds a distinct hepatic (delayed) regulation pathway beyond the insulin-effect latent in Benchmark 2.

Equations to recover.

$$\dot{G}_p(t) = EGP(t) + Ra(t) - U_{ii}(t) - E(t) - k_1 \cdot G_p(t) + k_2 \cdot G_t(t), \quad (47)$$

$$\dot{G}_t(t) = -U_{id}(t) + k_1 \cdot G_p(t) - k_2 \cdot G_t(t), \quad (48)$$

$$U_{id}(t) = \frac{[V_{m0} + V_{mx} \cdot X(t)] \cdot G_t(t)}{K_m + G_t(t)}, \quad (49)$$

$$U(t) = U_{ii}(t) + U_{id}(t), \quad (50)$$

$$\dot{X}(t) = -p_{2U} \cdot X(t) + p_{2U} \cdot (I(t) - I_b), \quad (51)$$

$$\dot{I}_1(t) = -k_i \cdot [I_1(t) - I(t)], \quad (52)$$

$$\dot{I}_d(t) = -k_i \cdot [I_d(t) - I_1(t)], \quad (53)$$

$$EGP(t) = k_{p1} - k_{p2} \cdot G_p(t) - k_{p3} \cdot I_d(t) - k_{p4} \cdot I_{po}(t). \quad (54)$$

Required unobservable variables. In these use cases, we require:

- glucose rate of appearance $Ra(t)$ to correctly attribute the post-meal glucose rise to the meal absorption process,
- an insulin-effect latent that mediates insulin-dependent disposal (in the ground truth model, this is the interstitial/remote insulin action state $X(t)$, which governs $U_{id}(t)$), and
- a delayed hepatic regulation latent that modulates endogenous glucose production. In the ground-truth model, this is the delayed insulin signal $I_d(t)$ realized by a two-compartment delay chain ($I_1(t)$, $I_d(t)$) and used in $EGP(t)$.

538 **Observables.** We observe the glucose concentration $G(t)$, the meal metadata (timing and amount),
539 and additionally insulin concentration $I(t)$. We may observe additional variables in the easy and
540 medium tiers.

- 541 • Easy: in addition to $G(t), I(t)$ and meal timing and amount, we observe all non-meal
542 contributions to $\dot{G}_p(t)$ and the major utilization terms needed to isolate insulin action,
543 namely $EGP(t), U_{ii}(t), E(t), G_t(t), U(t)$, and and portal insulin proxy $I_{po}(t)$.
- 544 • Medium: in addition to $G(t), I(t)$ and meal timing and amount, we observe $EGP(t)$.
- 545 • Hard: we observe no additional variables, because $G(t), I(t)$ and meal timing and amount
546 are the only ones routinely measured in clinical practices.

547 **Proposer Prompt.**

548 *A. Domain context.* You are building a mechanistic, analytic dynamical model for postprandial
549 glucose regulation. After a meal, glucose rises due to absorption, decreases due to insulin-mediated
550 disposal, and endogenous glucose production is suppressed via a delayed regulatory pathway.

551 *B. Requirements.*

- 552 • Propose a parsimonious continuous-time ODE model that can reproduce the observed
553 postprandial glucose trajectory given meal information, and insulin trajectory.
- 554 • Your model must be able to provide physiological interpretation that represents a delayed
555 regulatory effect on endogenous glucose production.
- 556 • The model must be interpretable: provide meaningful names for states and explain each
557 state’s role.
- 558 • The model must be analytic, namely closed-form expressions.

559 *C. Constraints and plausibility check.*

- 560 • Nonnegativity: physiological quantities and latent states should remain nonnegative.
- 561 • Meal causality: meal influence should start at/after meal time, not before.
- 562 • Transient response: meal effect should rise and then decay; no unrealistic oscillations.
- 563 • Basal plausibility: before meal, the system should be at/near steady state (or you must state
564 assumptions/initialization).

565 *D. Data columns available.*

- 566 • In the easy level, we have the glucose concentration $G(t)$, the insulin concentration $I(t)$,
567 the meal metadata (timing and amount), and $EGP(t), U_{ii}(t), E(t), G_t(t), I_{po}(t)$.
- 568 • In the medium level, we have the glucose concentration $G(t)$, the insulin concentration $I(t)$,
569 the meal metadata (timing and amount), and the endogenous glucose production $EGP(t)$.
- 570 • In the hard level, we have the glucose concentration $G(t)$, the insulin concentration $I(t)$,
571 and the meal metadata (timing and amount).

572 *E. Required output format.*

573 Return, in order:

- 574 • State list: name, unit (if known), interpretation.
- 575 • ODEs: one equation per state.
- 576 • Meal input encoding: how meal time/amount enters the model.
- 577 • Observation mapping: how predicted glucose is computed from states.
- 578 • Sanity-check procedure: how you verify nonnegativity, causality, and transient response.

579 **C.2.4 Benchmark 4: Flux Portrait Recovery under Partial Observability**

580 **Applications and use cases.** This benchmark targets scientific model discovery where the goal is not
581 only to fit concentrations but also to recover a coherent internal decomposition of the glucose–insulin
582 process. This corresponds to requiring the full multi-process flux portrait in addition to latent internal
583 states.

584 **Equations to recover.** All the equations in the ground truth model.

585 **Required unobservable variables.** In these use cases, we require the model to recover the multi-
586 process flux portrait:

- 587 • glucose rate of appearance $Ra(t)$,
- 588 • endogenous production $EGP(t)$,
- 589 • glucose utilization/disposal $U(t)$,
- 590 • insulin secretion driving insulin kinetics $S(t)$.

591 **Observables.** We observe the glucose concentration $G(t)$, the meal metadata (timing and amount),
592 and additionally insulin concentration $I(t)$. We may observe additional variables in the easy and
593 medium tiers.

- 594 • Easy: in addition to $G(t)$, $I(t)$ and meal timing and amount, we observe all the variables
595 except the unobservable ones.
- 596 • Medium: in addition to $G(t)$, $I(t)$ and meal timing and amount, we observe
597 $U_{ii}(t)$, $E(t)$, $G_t(t)$.
- 598 • Hard: we observe no additional variables, because $G(t)$, $I(t)$ and meal timing and amount
599 are the only ones routinely measured in clinical practices.

600 **Proposer Prompt.**

601 *A. Domain context.* You are building a mechanistic, analytic dynamical model for postprandial
602 glucose–insulin regulation whose purpose is both accurate trajectory prediction and mechanistic
603 interpretability. The model should provide a coherent decomposition of glucose dynamics into
604 appearance, endogenous production, utilization, and insulin secretion processes.

605 *B. Requirements.*

- 606 • Propose a parsimonious continuous-time ODE model that can reproduce the observed
607 postprandial glucose trajectory given meal information, and insulin trajectory.
- 608 • Your model must include unobserved latent mechanisms with physiological meaning that
609 (i) mediate meal-to-glucose influence over time, (ii) mediate insulin-to-disposal influence
610 over time, (iii) mediate delayed regulation of endogenous production, and (iv) drive insulin
611 secretion.
- 612 • The model must be interpretable: provide meaningful names for states and explain each
613 state’s role.
- 614 • The model must be analytic, namely closed-form expressions.

615 *C. Constraints and plausibility check.*

- 616 • Nonnegativity: physiological quantities and latent states should remain nonnegative.
- 617 • Meal causality: meal influence should start at/after meal time, not before.
- 618 • Transient response: meal effect should rise and then decay; no unrealistic oscillations.
- 619 • Basal plausibility: before meal, the system should be at/near steady state (or you must state
620 assumptions/initialization).

621 *D. Data columns available.*

- 622 • In the easy level, we have the glucose concentration $G(t)$, the insulin concentration $I(t)$,
623 the meal metadata (timing and amount), and all the variables except the unobservable ones.
- 624 • In the medium level, we have the glucose concentration $G(t)$, the insulin concentration $I(t)$,
625 the meal metadata (timing and amount), and $U_{ii}(t)$, $E(t)$, $G_t(t)$.
- 626 • In the hard level, we have the glucose concentration $G(t)$, the insulin concentration $I(t)$,
627 and the meal metadata (timing and amount).

628 *E. Required output format.*

629 Return, in order:

- 630 • State list: name, unit (if known), interpretation.
- 631 • ODEs: one equation per state.
- 632 • Meal input encoding: how meal time/amount enters the model.
- 633 • Observation mapping: how predicted glucose is computed from states.
- 634 • Sanity-check procedure: how you verify nonnegativity, causality, and transient response.

635 **C.3 Judge Prompt**

636 You are an evaluator for a dynamical system formulated based on text and data. You are given:

- 637 • The Proposer Prompt, and
- 638 • the candidate model (state list, ODEs, definitions of outputs, and assumptions).

639 Your job is to score how well the candidate satisfies the Proposer Prompt.

640 Step 1: Extract a checklist from the Proposer Prompt

641 From the Proposer Prompt, extract:

- 642 • Objective(s): what the model is meant to achieve.
- 643 • Required outputs: any signals the model must produce.
- 644 • Required latent concepts: any latent mechanisms/states the prompt demands.
- 645 • Constraints / plausibility checks: nonnegativity, causality, steady state, scaling, smoothness,
646 etc.
- 647 • Data availability: what columns are provided.

648 If the prompt is underspecified, list the ambiguities explicitly.

649 Step 2: Evaluate the Candidate Model (score 0–1 in each category)

650 Assign scores in $[0,1]$ with 1–3 sentence justification each:

- 651 • A. Output coverage: Does the model explicitly define all required outputs?
- 652 • B. Latent mechanism adequacy: Does it include at least one latent state with physiological
653 meaning that generates the required output?
- 654 • C. Constraint compliance: Does the candidate model respect physical constraints such as
655 nonnegativity, causality, and smoothness?
- 656 • D. Parsimony and interpretability: Is the model minimal while remaining interpretable? Are
657 state names meaningful?

658 Step 3: Aggregate score

659 Compute the average of the four scores above.

660 Step 4: Provide actionable feedback

661 Return:

- 662 • Top 3 missing items from the extracted checklist.

- Top 3 minimal edits that would improve the model.
- Any red flags (e.g., missing definition, negative rates, dependence on unavailable variables).

Output format (strict)

Return a JSON object with keys:

- extracted checklist
- category scores (A-D)
- average score
- feedback

C.4 Baseline Methods

We compare our strategy against three established symbolic regression approaches:

SINDy (Sparse Identification of Nonlinear Dynamics). SINDy [1] constructs a library of candidate functions and uses sparse regression (typically LASSO or sequentially thresholded least squares) to identify active terms. We use a polynomial library up to degree 2 with tanh basis functions where appropriate. SINDy is highly effective when the true dynamics lie within the predefined function library but cannot adapt its basis to novel functional forms.

PySR (Genetic Programming Symbolic Regression). PySR [2] evolves mathematical expressions using genetic programming with a multi-population island model. We configure it with binary operators $\{+, -, \times, /\}$ and unary operators $\{\tanh, \exp, \text{neg}\}$, running for 40 iterations per output variable. PySR can discover arbitrary compositions of operators but requires significant computation and may overfit without careful regularization.

D3 (Data-Driven Discovery). D3 [11] uses neural network-guided search with confidence interval-based model selection. We run D3 with its default configuration, which includes automatic complexity regularization. D3 balances expressiveness with interpretability through its probabilistic model selection framework.

C.5 Experimental Protocol

For each benchmark, we generate training and test trajectories from different random initial conditions. Specifically:

- **Training data:** Trajectory from seed s for coefficient fitting
- **Validation data:** Trajectory from seed $s + 1$ for model selection and pruning threshold optimization
- **Test data:** Trajectory from seed $s + 2$ for unbiased final evaluation

We run each method with 5 different random seeds (42–46) and report mean $\pm$ standard deviation of the test mean squared error (MSE). For our method, we use at most 50 LLM refinement iterations with the judge mechanism enabled for proposal validation. A threshold is fixed for the validation error, to terminate the LLM based optimization. The judge evaluates each proposed model structure for physical plausibility before coefficient fitting, reducing wasted computation on ill-formed hypotheses.

Benchmark	Tier	Method	Obs. MSE ↓	Spec. success ↑	Hidden MSE ↓	# Terms ↓
T1: Meal appearance	E	Ours	0.105 ± 0.148	1.0	0.179 ± 0.00	6.0 ± 2.8
		SINDy	0.050 ± 0.038	0.0	–	59.3 ± 2.9
		PySR	138.5 ± 239.4	0.0	–	61.3 ± 11.7
		D3	0.252 ± 0.206	0.0	–	33.7 ± 7.095
T1: Meal appearance	M	Ours	1.6e-04 ± 0.0e+00	1.0	0.180 ± 0.00	2.5 ± 0.7
		SINDy	0.470 ± 0.036	0.0	–	13.0 ± 0.00
		PySR	0.580 ± 0.053	0.0	–	26.0 ± 6.2
		D3	0.519 ± 0.135	0.0	–	52.3 ± 20.3
T1: Meal appearance	H	Ours	1.104 ± 1.561	1.0	0.505 ± 0.121	1.5 ± 0.7
		SINDy	2.172 ± 0.158	0.0	–	1.0 ± 0.0
		PySR	2.172 ± 0.158	0.0	–	11.7 ± 2.5
		D3	2.174 ± 0.159	0.0	–	34.0 ± 21.6
T2: Insulin action	E	Ours	0.190 ± 0.01	1.0	0.222 ± 0.018	11.0 ± 1.4
		SINDy	0.075 ± 0.101	0.0	–	136.7 ± 1.5
		PySR	0.462 ± 0.775	0.0	–	92.7 ± 3.1
		D3	0.033 ± 0.012	0.0	–	57.0 ± 12.5
T2: Insulin action	M	Ours	0.050 ± 0.00	1.0	0.617 ± 0.015	3.5 ± 0.7
		SINDy	0.024 ± 0.07	0.0	–	30.7 ± 3.215
		PySR	0.058 ± 0.010	0.0	–	42.0 ± 6.1
		D3	0.017 ± 0.01	0.0	–	57.3 ± 22.5
T2: Insulin action	H	Ours	0.108 ± 0.012	1.0	0.308 ± 0.00	2.5 ± 2.1
		SINDy	0.100 ± 0.012	0.0	–	8.0 ± 0.0
		PySR	0.049 ± 0.012	0.0	–	28.0 ± 2.0
		D3	0.032 ± 0.02	0.0	–	50.3 ± 11.0
T3: Hepatic delay	E	Ours	0.111 ± 0.101	1.0	0.594 ± 0.109	8.5 ± 2.1
		SINDy	1.699 ± 1.788	0.0	–	192.7 ± 0.6
		PySR	0.032 ± 0.014	0.0	–	118.3 ± 11.0
		D3	0.095 ± 0.053	0.0	–	60.3 ± 4.2
T3: Hepatic delay	M	Ours	0.066 ± 0.00	1.0	0.671 ± 0.111	2.5 ± 2.1
		SINDy	0.015 ± 0.02	0.0	–	18.0 ± 0.0
		PySR	0.041 ± 0.03	0.0	–	50.0 ± 4.583
		D3	0.049 ± 0.048	0.0	–	48.7 ± 8.083
T3: Hepatic delay	H	Ours	0.100 ± 0.00	1.0	0.499 ± 0.061	6.0 ± 4.2
		SINDy	0.100 ± 0.012	0.0	–	8.0 ± 0.0
		PySR	0.054 ± 0.020	0.0	–	26.0 ± 7.2
		D3	0.030 ± 0.05	0.0	–	25.3 ± 7.6
T4: Flux portrait	E	Ours	0.010 ± 0.014	1.0	0.324 ± 0.295	27.0 ± 9.9
		SINDy	1.8e+04 ± 1.1e+03	0.0	–	203.7 ± 26.6
		PySR	260.1 ± 439.3	0.0	–	219.3 ± 18.8
		D3	0.720 ± 0.475	0.0	–	37.7 ± 2.5
T4: Flux portrait	M	Ours	0.026 ± 0.037	1.0	0.412 ± 0.218	5.5 ± 2.1
		SINDy	0.039 ± 0.030	0.0	–	53.3 ± 1.528
		PySR	0.659 ± 1.088	0.0	–	65.7 ± 15.3
		D3	0.107 ± 0.098	0.0	–	63.0 ± 15.4
T4: Flux portrait	H	Ours	0.197 ± 0.278	1.0	0.464 ± 0.067	5.0 ± 4.2
		SINDy	0.100 ± 0.012	0.0	–	8.0 ± 0.0
		PySR	0.055 ± 0.07	0.0	–	28.3 ± 4.2
		D3	0.216 ± 0.323	0.0	–	38.7 ± 14.8

Table 4: Main benchmark results across 12 benchmarks.

699 D Main Results on All 12 Benchmarks

E Details on Case Study

A representative discovered model constructs the following task-aligned latent surrogates from the observed glucose–insulin trajectories:

$$\begin{aligned} z_{\text{act}}(t) &= \frac{I(t)^2}{1 + I(t)^2 + \dot{I}(t)^2}, \\ z_{\text{meal}}(t) &= \frac{\dot{G}_p(t)^2}{1 + \dot{G}_p(t)^2}, \\ z_{\text{hep}}(t) &= \frac{z_{\text{act}}(t) + 10^{-3}z_{\text{meal}}(t)}{1 + z_{\text{act}}(t) + 10^{-3}z_{\text{meal}}(t)}, \\ z_{\text{egp}}(t) &= \frac{1}{1 + z_{\text{hep}}(t) + 10^{-3}G_p(t)^2}, \\ z_{\text{util}}(t) &= \frac{z_{\text{act}}(t)G_p(t)}{1 + G_p(t)}, \\ z_{\text{sec}}(t) &= \frac{G_p(t)}{1 + G_p(t)} + 10^{-3} \frac{\dot{G}_p(t)^2}{1 + \dot{G}_p(t)^2}. \end{aligned}$$

These variables are not intended to exactly reproduce the original Dalla Man state labels or units. Rather, they form an interpretable reduced state representation: z_{meal} tracks meal-driven appearance, z_{act} tracks insulin action, z_{hep} and z_{egp} encode hepatic suppression and production, z_{util} encodes insulin-dependent utilization, and z_{sec} encodes glucose-driven secretion. The graph meta-model then uses these process variables in a linear-in-parameters glucose–insulin balance:

$$\begin{aligned} \dot{G}_p(t) &= \theta_1 z_{\text{meal}}(t) + \theta_2 z_{\text{egp}}(t) + \theta_3 z_{\text{util}}(t) + \theta_4 z_{\text{basal}}(t) + \theta_0, \\ \dot{I}(t) &= \theta_5 z_{\text{sec}}(t) + \theta_6 z_{\text{clear}}(t) + \theta_7, \end{aligned}$$

where $z_{\text{basal}}(t) = G_p(t)/(1 + G_p(t))$ and $z_{\text{clear}}(t) = I(t)^2/(1 + I(t)^2)$. This structure matches the qualitative form required by the task: glucose dynamics are decomposed into appearance, production, and utilization terms, while insulin dynamics are decomposed into secretion and clearance terms.

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978 submissions via structured templates. This includes details about training, license,
979 limitations, etc.

980 • The paper should discuss whether and how consent was obtained from people whose
981 asset is used.

982 • At submission time, remember to anonymize your assets (if applicable). You can either
983 create an anonymized URL or include an anonymized zip file.

984 **14. Crowdsourcing and research with human subjects**

985 Question: For crowdsourcing experiments and research with human subjects, does the paper
986 include the full text of instructions given to participants and screenshots, if applicable, as
987 well as details about compensation (if any)?

988 Answer: [N/A]

989 Justification: No humans involved

990 Guidelines:

991 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
992 with human subjects.

993 • Including this information in the supplemental material is fine, but if the main contribu-
994 tion of the paper involves human subjects, then as much detail as possible should be
995 included in the main paper.

996 • According to the NeurIPS Code of Ethics, workers involved in data collection, curation,
997 or other labor should be paid at least the minimum wage in the country of the data
998 collector.

999 **15. Institutional review board (IRB) approvals or equivalent for research with human**
1000 **subjects**

1001 Question: Does the paper describe potential risks incurred by study participants, whether
1002 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
1003 approvals (or an equivalent approval/review based on the requirements of your country or
1004 institution) were obtained?

1005 Answer: [N/A]

1006 Justification: No IRB needed

1007 Guidelines:

1008 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
1009 with human subjects.

1010 • Depending on the country in which research is conducted, IRB approval (or equivalent)
1011 may be required for any human subjects research. If you obtained IRB approval, you
1012 should clearly state this in the paper.

1013 • We recognize that the procedures for this may vary significantly between institutions
1014 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
1015 guidelines for their institution.

1016 • For initial submissions, do not include any information that would break anonymity (if
1017 applicable), such as the institution conducting the review.

1018 **16. Declaration of LLM usage**

1019 Question: Does the paper describe the usage of LLMs if it is an important, original, or
1020 non-standard component of the core methods in this research? Note that if the LLM is used
1021 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
1022 scientific rigor, or originality of the research, declaration is not required.

1023 Answer: [N/A]

1024 Justification: LLM not significant in the core methods etc.

1025 Guidelines:

- 1026 • The answer [N/A] means that the core method development in this research does not
- 1027 involve LLMs as any important, original, or non-standard components.
- 1028 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
- 1029 be described.